

Politecnico di Torino

Passive Optical Components
Assignments Report

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Chapter 1

Assignement 5: Micro-Ring Resonators

1.1 Introduction and Objectives

The main objective of this assignment is to theoretically model and numerically simulate the spectral behavior of an MRR relying on the scattering matrix (S-matrix) formalism. Instead of using standard coupled-mode differential equations, the device is modeled by treating the coupling region as a four-port directional coupler, where one of the output ports is directly fed back into an input port to form the closed resonant loop.

During this analysis, the following key steps were implemented and evaluated in MATLAB:

- Design and simulation of a single-bus Micro-Ring Resonator, evaluating its spectral transmission and input reflectivity (return loss).
- Introduction of a realistic non-ideal coupling condition. Specifically, the isolation parameter of the scattering matrix (S_{41}) was set to a small but non-zero value, which is strictly required to properly observe and quantify the return loss at the input port.
- Implementation of a custom function to systematically evaluate the device under different physical conditions, comparing multiple examples (varying ring radius and coupling coefficient).
- Extension of the mathematical model to an Optical Add-Drop Multiplexer configuration, in which the resonant ring is coupled between two separate bus waveguides.
- Extraction and numerical evaluation of the core performance parameters of the device, such as the Free Spectral Range (FSR), the Full Width at Half Maximum (FWHM), and the Quality Factor (Q).

1.2 Mathematical Model of the Single Micro-Ring Resonator

To properly describe the spectral response of a single-bus Micro-Ring Resonator (MRR), the system is modeled by representing the coupling region as a four-port device charac-

terized by a 4×4 scattering matrix. The resonant cavity is then mathematically formed by connecting one of the output ports back to an input port, forcing the optical field to circulate within the loop.

In a standard directional coupler, the S-matrix relates the complex amplitudes of the incoming waves to the outgoing ones. Let C represent the cross-coupling amplitude and t the straight-through transmission coefficient. However, an ideal, perfectly symmetric directional coupler exhibits zero reflection at the isolated port, meaning no light would be back-reflected into the input waveguide. In this analysys, a physical device must be considered, which means the matrix element S_{41} must be realistically set as small and non zero value.

To respect the conservation of energy within the coupling region, the inclusion of the non ideal back reflection magnitude ($S_{41,mag}$) requires a recalculation of the direct transmission coefficient t. The elements of the scattering matrix are therefore defined as:

$$S_{21} = t = \sqrt{1 - C^2 - S_{41,mag}^2} \tag{1.1}$$

$$S_{31} = S_{42} = jC \tag{1.2}$$

$$S_{41} = S_{41,mag} \tag{1.3}$$

Once the light couples into the ring, it travels along the circumference $L = 2\pi r$, accumulating a phase shift θ. Assuming a weakly guiding waveguide with an effective refractive index n_{eff}, the phase shift depends on the operating wavelength λ as:

$$\theta = \beta L = \frac{2\pi n_{eff}}{\lambda} L \tag{1.4}$$

where β is the modal propagation constant. The propagation inside the loop is fully described by a feedback term that accounts for both the phase accumulation and the optical attenuation. Defining α as the loss coefficient, the round-trip amplitude attenuation factor is $a = \exp(-\alpha L/2)$. The overall feedback term is thus expressed as $ae^{-j\theta}$.

By applying the boundary conditions dictated by the closed loop, the output fields can be derived algebraically from the S-matrix. The resulting equation for the input reflectivity, which represents the Return Loss of the device (Γ_{in}), is obtained by solving the multi-port interference:

$$\Gamma_{in} = \frac{2S_{41}S_{42}ae^{-j\theta}}{1 - S_{21}ae^{-j\theta}} \tag{1.5}$$

Similarly, the complex amplitude of the light passing through the straight bus waveguide (Transmission, T) is derived by summing the directly transmitted field with the infinite contributions of the light coupling back from the ring:

$$T = S_{21} + \frac{(S_{31}^2 + S_{41}^2)ae^{-j\theta}}{1 - S_{21}ae^{-j\theta}} \tag{1.6}$$

Finally, the physical observables of the system are the normalized optical powers, which are calculated as the squared magnitudes of the respective complex fields:

$$R_{power} = |\Gamma_{in}|^2 \quad \text{and} \quad T_{power} = |T|^2 \tag{1.7}$$

In order to simulate the behavior of the MMR with different parameters in an easier way, the computation of the T and R parameters has been defined in a local function.

A baseline case was defined with a ring radius $r = 8\,\mu$m, an effective refractive index $n_{eff} = 2.6$, and an ideal lossless propagation condition ($a = 1$). The power coupling coefficient was set to $K = C^2 = 0.10$, while the non-ideal reflection term was fixed at $S_{41,mag} = 0.05$. Additional examples were generated by varying both the coupling strength and the ring radius.

1.3 Spectral Response of the Single MRR

The numerical simulation was performed over a wavelength span centered around the typical optical communication band, specifically focusing on the region near $1.555\,\mu$m.

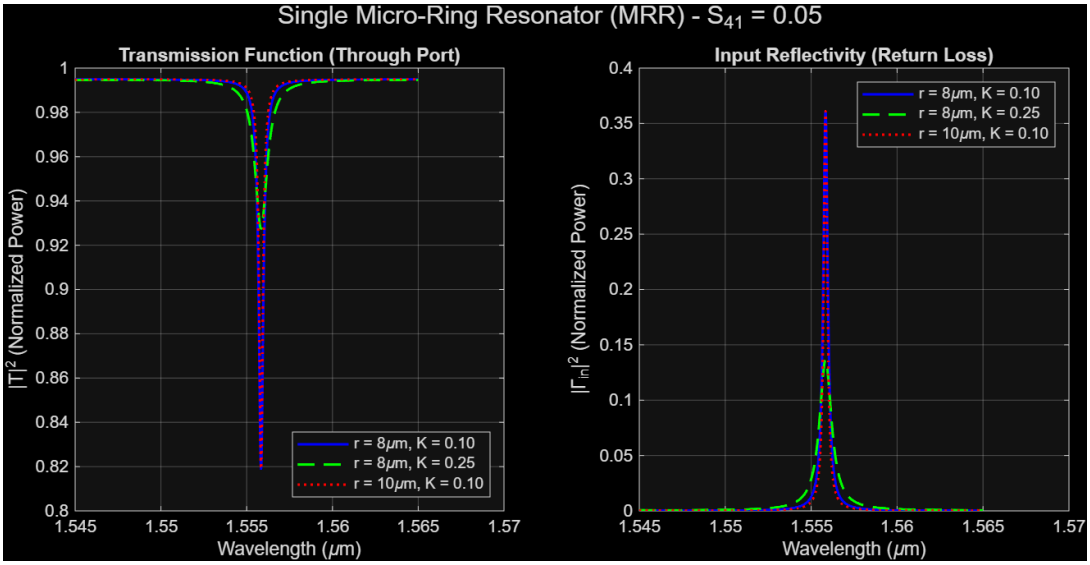


Figure 1.1: Spectral response of the single Micro-Ring Resonator (MRR) comparing different coupling coefficients and ring radiuses. The left panel shows the normalized transmission power (Through port) behaving as a notch filter. The right panel displays the input reflectivity (Return Loss) generated by the non-ideal coupling parameter S_{41}.

As illustrated in Figure 1.1 (left panel), the transmission spectrum exhibits the characteristic optical behavior of a notch filter. For wavelengths that do not satisfy the phase-matching condition, the light effectively bypasses the ring and travels along the straight bus waveguide undisturbed, resulting in a normalized transmission power close to unity. However, when the wavelength perfectly matches the resonance condition of the cavity ($Ln_{eff} = m\lambda_{res}$), the optical power couples into the ring and experiences a significant build-up. The light coupling back from the ring into the bus waveguide destructively interferes with the incoming straight-through wave, leading to a drop in the transmitted power.

Moreover, the introduction of the non ideal scattering parameter $S_{41,mag} = 0.05$ allows the parameter Return Loss not to be null, as shown in the right panel of Figure 1.1. At resonance, the high-intensity optical field circulating inside the loop interacts with the non-ideal coupling region, leaking power back into the isolated input port. This phenomenon generates a reflection peak located at the resonance wavelength.

To observe the parametric dependence of the MRR, multiple configurations were compared:

- **Varying the coupling coefficient (K):** By increasing the power coupling from $K = 0.10$ to $K = 0.25$ (green dashed line), the resonance peak becomes visibly broader. This occurs because the quality factor Q get lower, due to higher FWHM (light exits faster from the ring).

- **Varying the ring radius (r):** By increasing the ring radius from $r = 8\,\mu$m to $r = 10\,\mu$m (red dotted line), the phase-matching condition is altered. This causes a clear shift in the resonance wavelength and a reduction in the Free Spectral Range (FSR), as the larger cavity supports more closely spaced resonant modes.

1.4 Extension to the Optical Add-Drop Multiplexer (OADM)

To enhance the functional capabilities of the single-bus MRR, the analysis was extended to an Optical Add-Drop Multiplexer (OADM) device. This configuration involves a resonant ring sandwiched between two parallel straight waveguides, enabling multi-port operation (Input, Through, Drop, and Add).

The OADM is treated by cascading the scattering matrices of two separate directional couplers, characterized by their respective power coupling coefficients K_1 and K_2. To focus on the ideal spectral filtering properties of the Add-Drop configuration and reproduce the reference curves, the non-ideal reflection parameter (S_{41}) was neglected in this step. Therefore, the straight-through transmission and cross-coupling amplitudes for the two couplers ($i = 1, 2$) are defined purely by their coupling strength:

$$t_i = \sqrt{1 - K_i}, \quad C_i = \sqrt{K_i} \tag{1.8}$$

In steady-state conditions, the optical field injected at the input port couples into the ring through the first coupler (K_1) and circulates. However, unlike the single-bus configuration, a fraction of the circulating energy is continuously extracted by the second coupler (K_2) into the Drop port. The overall feedback term for a complete round-trip becomes $t_1 t_2 a e^{-j\theta}$, as the circulating field is diminished by the transmission coefficients of both couplers.

Solving the scattering matrix equations gives the electric field at the Through port (E_{trans}). This field comes from the interference between the light traveling straight through the bus and the fraction circulating inside the ring that couples back into it:

$$E_{trans} = \frac{t_1 - t_2 a e^{-j\theta}}{1 - t_1 t_2 a e^{-j\theta}} \tag{1.9}$$

For the Drop port, the coupled light must travel along exactly half of the ring's circumference ($L/2$) before encountering the second coupler. The field accumulates half of the total phase shift ($\theta/2$) and half of the total round-trip attenuation ($\sqrt{a}$). Taking into account the cross-coupling coefficients of both junctions, the extracted complex field (E_{drop}) is formulated as:

$$E_{drop} = \frac{-C_1 C_2 \sqrt{a} e^{-j\frac{\theta}{2}}}{1 - t_1 t_2 a e^{-j\theta}} \tag{1.10}$$

Finally, the normalized optical intensities corresponding to the transmission (I_{trans}/I_{in}) and drop (I_{drop}/I_{in}) spectra are evaluated by computing the squared magnitudes of the

respective complex fields:

$$P_{trans} = |E_{trans}|^2 \quad \text{and} \quad P_{drop} = |E_{drop}|^2 \tag{1.11}$$

1.5 OADM Results and Performance Parameters

The OADM model was again simulated in MATLAB comparing two symmetric coupling configurations: a baseline design with $K_1 = K_2 = 0.10$ and a higher-coupling design with $K_1 = K_2 = 0.20$. The resulting spectral responses for the Through and Drop ports are presented in Figure 1.2.

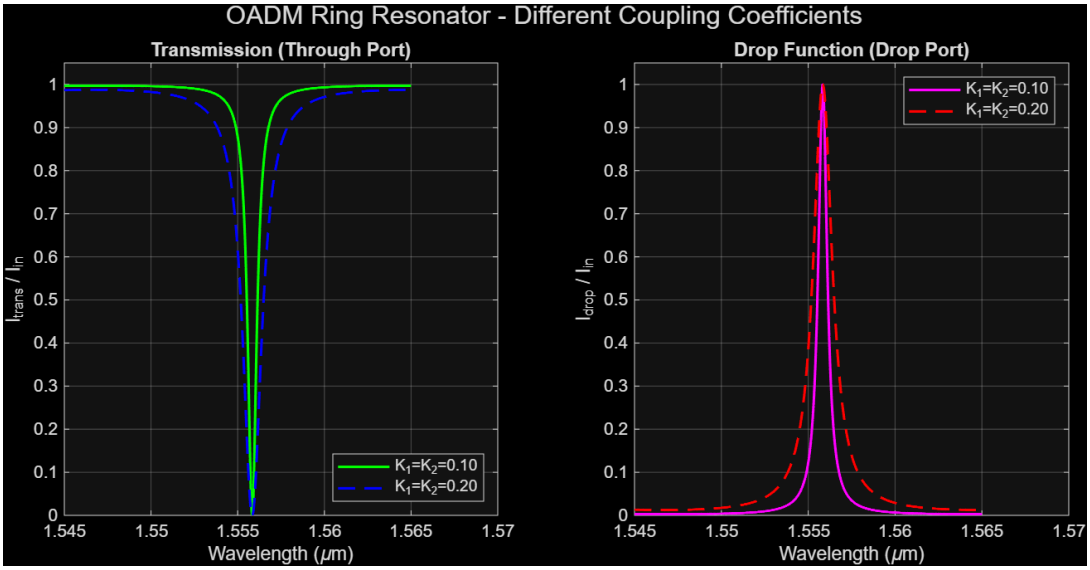


Figure 1.2: Spectral response of the Optical Add-Drop Multiplexer (OADM) configuration comparing different coupling strengths. The left panel shows the transmission at the Through port, while the right panel displays the extracted signal at the Drop port.

The numerical results are as expected. The Transmission spectrum (Through port) acts as a highly selective notch filter as before, this time dropping to zero exactly at the resonance wavelength due to the symmetric coupling condition, which guarantees total destructive interference for the straight-through wave. Instead, the Drop port acts as a narrow bandpass filter: at resonance, the optical power circulating inside the ring is totally extracted by the second coupler, resulting in a sharp transmission peak. Off-resonance, the signal bypasses the ring entirely, leaving the Drop port perfectly isolated.

As observed in Figure 1.2, increasing the coupling coefficients (from 0.10 to 0.20) broadens the passband at the Drop port, highlighting the intrinsic trade-off between spectral selectivity and coupling strength.

To fully characterize the optical performance of the designed MRR, three parameters were evaluated from the simulated data for the baseline case ($r = 8\,\mu$m, $K_1 = K_2 = 0.10$): the Free Spectral Range (FSR), the Full Width at Half Maximum (FWHM), and the Quality Factor (Q).

The **Free Spectral Range (FSR)** represents the spectral distance between two adjacent resonance peaks. For a weakly dispersive waveguide, it is analytically approximated as:

$$FSR \approx \frac{\lambda_{res}^2}{n_{eff} L} \tag{1.12}$$

Given the design parameters and the extracted resonance wavelength $\lambda_{res} = 1555.835$ nm (approximately $1.556\,\mu$m), the calculated FSR is 18.52 nm. This relatively large FSR explains the presence of a single resonance peak within the simulated 20 nm window.

The **Full Width at Half Maximum (FWHM)** defines the 3-dB bandwidth of the resonance peak and dictates the spectral resolution of the filter. From the simulated drop-port spectrum, the extracted FWHM value is 621.46 pm.

Finally, the **Quality Factor (Q)** is a dimensionless parameter that measures the sharpness of the resonance, defined as the ratio between the operating wavelength and the resonance width:

$$Q = \frac{\lambda_{res}}{FWHM} \tag{1.13}$$

The obtained Quality Factor is $Q = 2504$. This high value confirms the strong confinement of the optical field within the ring and demonstrates the device's suitability for high-resolution wavelength filtering and label-free sensing applications.

The complete set of performance parameters extracted from the simulation of the baseline case is summarized in Table 1.1.

Table 1.1: Extracted performance parameters for the baseline OADM configuration ($r = 8\,\mu$m, $K = 0.10$).

Parameter	**Symbol**	**Simulated Value**
Resonance Wavelength	λ_{res}	1555.835 nm
Free Spectral Range	FSR	18.52 nm
Full Width at Half Maximum	$FWHM$	621.46 pm
Quality Factor	Q	2504